Program: **Diploma in Automation and Robotics**	
Course Code: **6339A**	Course Title: **Advanced Robotics Lab**
Semester: **6**	Credits: **1.5**
Course Category: **Program Elective**	
Periods per week: **3 (L:0 T:0 P:3)**	Periods per semester: **45**

Course Objectives:

- To introduce the concepts of mobile robots and its navigation
- Familiarize with the robot programming and control

General Instructions:

- Students have to write the date, aim, Software & Hardware requirements for that experiment in the observation book.
- Students have to listen and understand the experiment explained by the faculty and note down the important points in the observation book.
- Students need to write procedure/algorithm in the observation book.
- Analyze and Develop/implement the logic of the program by the student in respective platform
- After approval of logic of the experiment by the faculty then the experiment has to be executed on the system.
- After successful execution the results are to be shown to the faculty and noted the same in the observation book.
- Students need to attend the Viva-Voce on that experiment and write the same in the observation book.
- Update the completed experiment in the record and submit to the concerned faculty in-charge.

Course Prerequisites:

Topic	Course code	Coursename	Semester
Basics of Robotics	2371	Foundations of Robotics	2

Tools and Equipment:

Robotic Manipulator
Processors: Intel Atom® processor or Intel® Core™ i3 processor.
Disk space: 1 GB.
Operating systems: Windows 7 or later, macOS, and Linux.
Software: Motosim or other Robotic Simulator

Course Outcomes:

On completion of the course, the student will be able to:

Con	Description	Duration (Hours)	Cognitivelevel
CO1	To familiarize with motion control of mobile robots.	9	Applying
CO2	To apply various path planning algorithm for mobile robots.	12	Applying
CO3	To assemble industrial robotic arm	12	Applying
CO4	To demonstrate various operations using teach pendant and simulate the robot motion using RoboX	9	Applying
	Lab Exam	3	

CO–POMapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3				3		
CO2	3						
CO3	3			3			
CO4	3			3			
CO5	3			3			

3-Strongly mapped,2-Moderately mapped,1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**To familiarize with motion control of mobile robots**		

M1.01	White line follower using Firebird V robot.	3	Applying
M1.02	Mobile controlled robot using Firebird V	3	Applying
M1.03	Gripper control using Firebird V	3	
CO2	**To apply various path planning algorithm for mobile robots.**		
M2.01	Path Traversal Algorithm using Firebird V Robot	6	Applying
M2.02	Slam algorithm on Firebird V	6	Applying
	Lab ExamI	1.5	
CO3	**To assemble industrial robotic arm**		
M3.01	Testing and Servo setting of servo motors	3	Applying
M3.02	Assembly of Arm	3	Applying
M3.03	Arduino Programming for Robotic Arm	3	Applying
M3.04	Controlling Robotic Arm using blue tooth	3	Applying
CO4	**To demonstrate various operations using teach pendant and simulate the robot motion using RoboX**		
M4.01	Robot jogging and understanding coordinate systems using teach pendant	3	Applying
M4.02	Teaching points to the robot using teach pendant	3	Applying
M4.03	Simulation of robot using RoboX software	3	
	LabExam II	1.5	

Online Resources:

Sl.No	Website Link/Textbooks
1	Fire Bird V ATMEGA2560 Hardware Manual
2	http://elsi.e-yantra.org/resources
3	RoboXsoftware Manual
4	https://github.com/